

The background of the slide is a vibrant, futuristic cityscape at night. Tall skyscrapers are illuminated with orange and yellow lights, and their forms are partially obscured by large, billowing white clouds. Overlaid on the scene are numerous vertical and horizontal lines in shades of blue and green, resembling digital data streams or fiber optic cables. The overall atmosphere is high-tech and dynamic.

Resiliency Assurance Service PSS

Last Updated: 24-Feb-26 | Public

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Resiliency Assurance Service

Resilience is a state of preparedness and capabilities that are achieved, in part, through strong orchestration of the emergency response, crisis management, IT disaster recovery and business continuity disciplines.

The major key elements and steps to achieve resilience are.

- **EMERGENCY RESPONSE:** The immediate reaction and response to an incident and reducing the overall severity and impact of the event.
- **CRISIS MANAGEMENT:** The overall coordination of an organization's response to a crisis level event in an effective, timely manner, focused on high-impact decision making and limiting the organization's profitability, reputation, and operational losses.
- **BUSINESS CONTINUITY:** Restore business operations to an acceptable level of care and performance within a defined timeframe following a disruption.
- **DISASTER RECOVERY:** Infrastructure and procedures required to restore critical infrastructure, applications, and data in line with business expectations for recovery

Business Resiliency requires a holistic approach and needs all 4 aspects to be part of this as inclusive and integrated. This PSS and offering only focus on the Disaster Resilience (Disaster recovery) scope and is limited.

Yotta's Resiliency Assurance Services combines proven business continuity and disaster recovery methodologies With cloud-based technologies to deliver end-to-end enterprise resiliency across hybrid, multiplatform IT Environments are cost-effective and reliable.

How Service Works

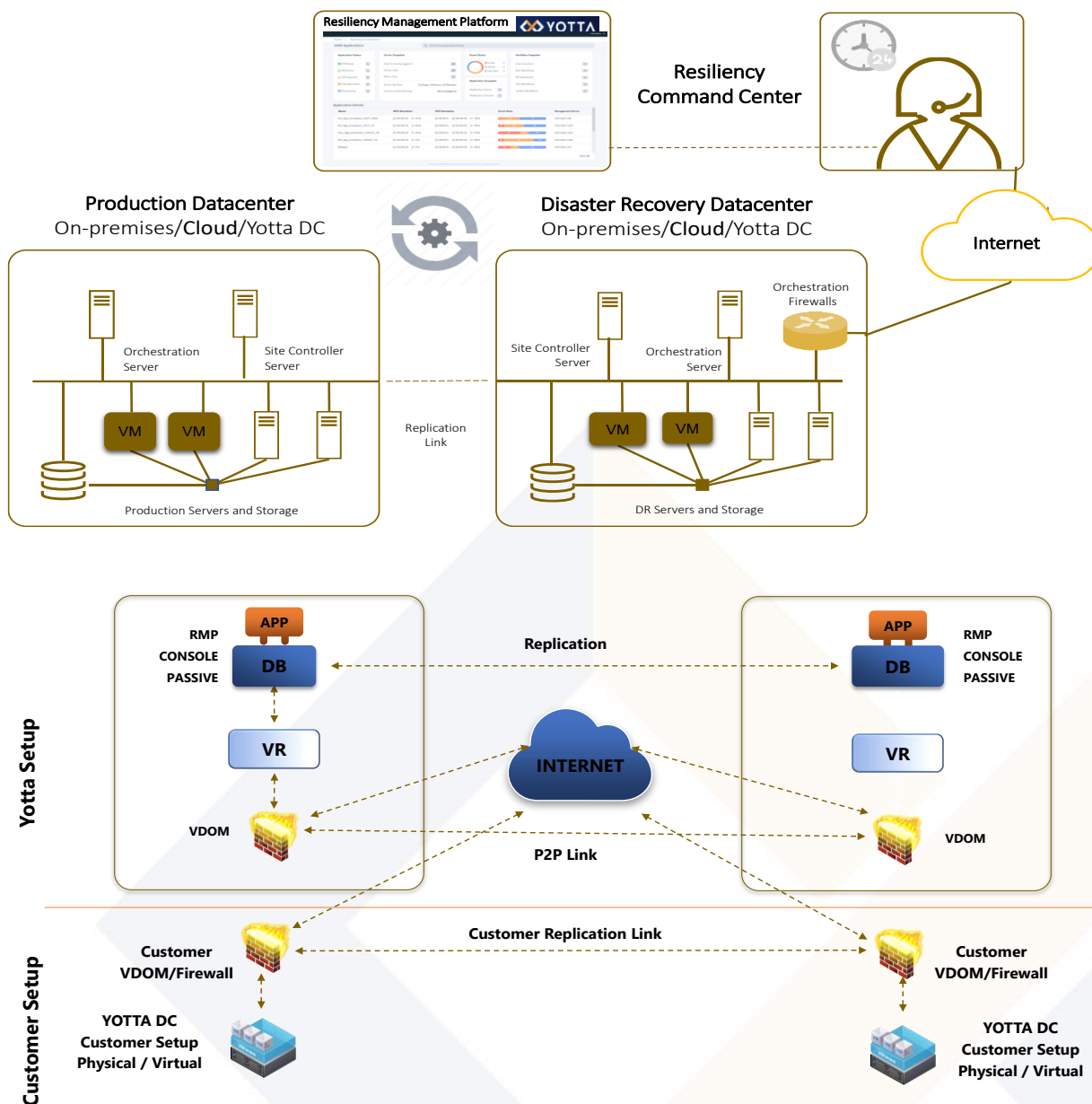
Yotta Resiliency Assurance service offers workflow-based automation and orchestration service that helps Customers recover their systems and resume back to normal operation. Further Yotta offer managed and/or assisted service to the customer for a seamless failover process, ensuring users are redirected from the primary environment to the DR environment. Service also monitors disaster recovery operations and helps customers recover systems and resume normal operation.

Yotta Resiliency Assurance service is especially useful for businesses that lack in-house experts to design and Execute disaster recovery plans. The ability to outsource infrastructure is another benefit for smaller organizations because it avoids the high cost of equipment needed to run a disaster recovery site.

IT Resiliency Key Considerations

- Resiliency or DR Site analysis and selection
- Identification of critical data and applications
- Resilience or DR Plans
- IT infrastructure like servers, network, storage etc.
- Mapping interdependencies
- IT Recovery Process and DR Automation
- Data mover and replication technology
- RPO and RTO requirements
- Testing and validation
- DRILL frequency and DR Management

Deployed Architecture



Service Components

Enterprises have their Data Centre's distributed across the geographical spread of their businesses and further opt for hybrid environments to accommodate pre-existent legacy systems in the new-age cloud environment. Looking at this new reality, it is more important than both DC-DR, as well as hybrid infra, are managed by efficient and effective resiliency service.

Before you subscribe to Yotta's Resiliency Assurance service few subsystems and strategy need to be prepared or needs to be defined by the organization, and those are.

- Nature and criticality of applications catering to different business services.
- Classify them as per their criticality (example: Low, Medium, and High)
- Servers or IT equipment environments and databases which are to be considered.
- The network that provides access and security to information within the intranet, from the Internet and VPNs
- Disk storage and frequency of access of these storage systems for different applications and services
- What is your RPO and RTO expectation and how much business can accommodate.
- Based above systems and plans please plan your DR IT infra planning.
 - ✓ Identify applications that need to be considered for DR or resiliency service.
 - ✓ Number of Servers to be protected.
 - ✓ CPU/RAM and storage capacity of the recovery server (at DR Site)
 - ✓ Amount of data to be replicated / continuous movement from the primary site.
 - ✓ Number of DR drills in a year or any frequency as per your requirement
 - ✓ Service Scenario of DR, RPO ~X hours and RTO ~Y hours

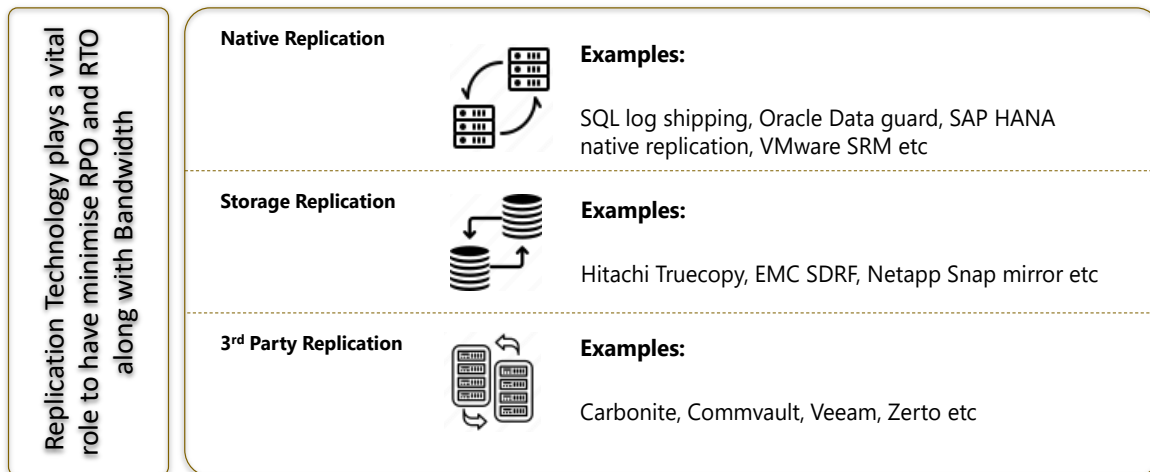
Major components of the Disaster resiliency assurance service are illustrated below.

Resiliency Orchestration: Resiliency Orchestration is a framework-based continuity management software that enables consistent operational interfaces and reduces the expertise required to configure and operate the application continuity and disaster recovery solutions. It encapsulates all the required intelligence – processes and procedures required to interface with applications, databases, replication mechanisms, servers, storage, and networks to automate deployment and operation of the continuity solution.

Resiliency Management Platform: It offers a single Resiliency management console view of enterprise IT continuity health. It interfaces with the various layers of the continuity solutions and reports deviation on Recovery Point Objective (RPO) and Recovery Time Objective (RTO) of the application, in real-time with Maximum Tolerable Period Destruction (MTPD).

Replication / Data mover Tools: Replication can be extended by 3rd party tool or customers can look at Native replication tools like SQL log shipping, Oracle data guard, etc. Furthermore, native can be integrated and monitored through the RMP portal. Yotta can extend the CARBONITE replication tool for file and block-level replication. Which is further integrated and available under Resiliency Assurance Service.

Some examples of selection mechanisms for replication technology.



Please note: This is just an example of selection and suggestion to have a choice of replication technology. The Same can be revalidated and checked the feasibility while the proposal stage or deployment.

Switch over and Switch Back / Failover and Fail Back

- Switchover and Switchback:** The Switchover is a **planned activity** to move to the DR site by following all the required processes and involving the appropriate stakeholders. The Switch over to the DR site infrastructure is done using the Yotta RAS orchestration tool and by using the deployed underline replication technology. Once the Switchover is done, we initiate the reverse replication which happens from DR to Production site to make sure that all the data is replicated and hence no data loss is possible during testing.
- In case of Switchback, we use the same steps from DR site which can also be called as the current Production site. Hence it is called Switchback. Once the Switchback is done normal replication is initiated. This will be done using the Yotta RAS management console. After the environment and data re-synchronization (i.e., ensuring all the environments are in synch), the systems will be brought back up at the Production site and made ready for operations to function from the Production site.
- Failover and Failback:** A failover is usually an **unplanned activity**, which occurs when the Production site experiences a disaster situation. A disaster situation could be any situation, which renders the Production site to become unusable for an extended period. An executive decision is typically taken, after a review of the situation (to be considered a disaster) is done with experts, to declare a disaster and to fail over the Production environment to the DR site. Once the decision is taken to fail over, the Production site is failed over to the DR environment. The Failover to the DR site infrastructure is done using Yotta RAS (as per the defined RPO and RTO). For known scenarios that are defined as part of DR operations, no significant data or as expected by the SLA is expected.

Failback operations are a set of planned activities to move from the DR site to the Production site where the original roles of the Production and DR environments are restored. This will include the following:

- Restoration of the Production environment could include procurement and deployment of new HW etc., restoration of data using backups, etc. from the DR site. This is like the setup of a DR site.
- Re-synchronizing the DR and the Production environments using replication technologies. This will be done using Yotta RAS like how it will be done for the DR setup. This will include reverse replication from the DR to the Production environment.
- Once the Production and the DR environments are in synch as required, the Failover from DR to the production site can be initiated as a planned activity, which is usually like switchback activities. This will be done using Yotta RAS. Systems will be brought back up at the Production site and made ready for operations to function.

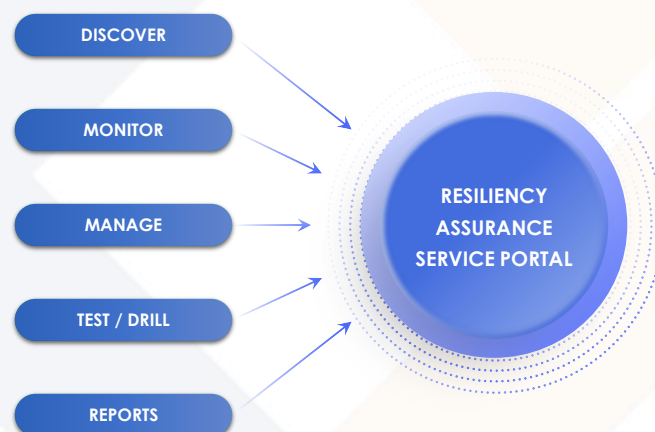
Application Workflow: Workflow will be custom stitched with Yotta RAS dynamic workflow creator. This workflow Will be used to create an application recovery procedure at the DR site.

Network Solution for DR Failover: DNS changes required at the DR site can also be automated using Yotta RAS. We will perform network-related changes at the DR site, and we will ensure that the same is available in RTO time or less from the time disaster is declared by Management.

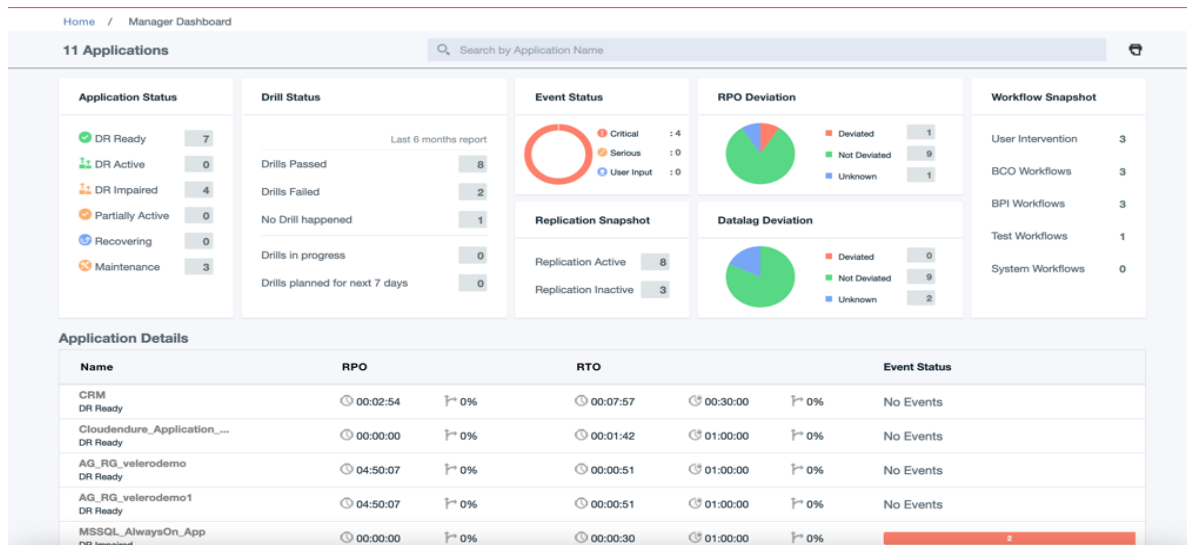
Please note: Global DNS automation is only possible at the IPS level.

RAS Management Portal

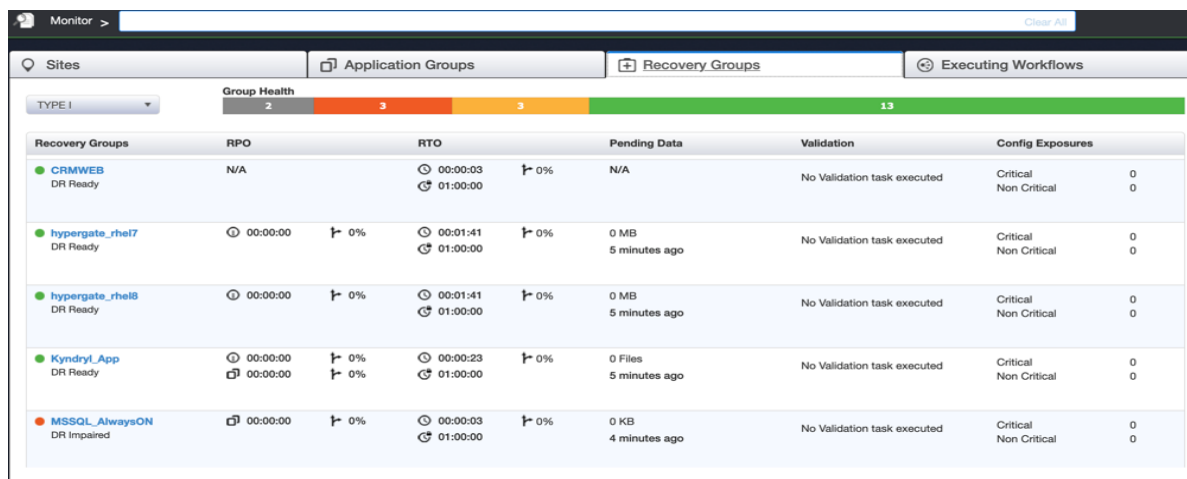
Yotta **Resiliency Assurance Service:** End-to-end resiliency management and supports almost “all” parameters which are illustrated here and below.



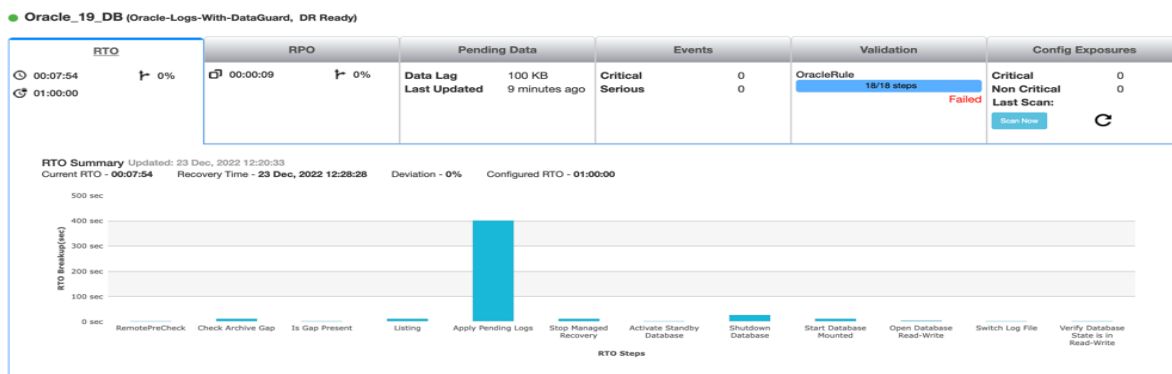
1. EXECUTIVE MANAGEMENT VIEW



2. IT TEAM / ADMINSTATOR MONITORING VIEW



3. RPO AND RTO MONITORING



With the Yotta Resiliency assurance service, enterprises have the flexibility to enable automation between different models of Datacenter setups. This service includes

- Implementation and integration of DR automation platform (RMP)
- DR Drill or Test
- Replication, monitoring along with RPO & RTO
- DR monitoring and management

Along with a subscription to resiliency assurance service other infrastructure components either customer needs to provide or subscribe as additional services, which are listed below.

- Yotta Public cloud (YEC)
- Private Cloud or hybrid cloud
- 3rd party Data mover or Replication tools/services
- Yotta Colocation and support services
- Network and Connectivity Services
- ✓ Public Internet / Private connectivity
- Cyber security services
- IT Management Services
- Any other tech or IT services

In addition to these components, the customer may subscribe to other managed services such as Security Services and Offsite Backup services. More information on these services is detailed in their respective service PSS or connect with our relationship manager or for more information, please refer or visit this URL:

<https://www.yotta.com/>

Flexible and Multiple Options of Resiliency / DR Site

As we all know, today businesses and its Data Centres are distributed across the globe and hybrid and multi-cloud environments is the reality and more prevalent times to come. So, there may be the situation that DC and DR sites and Resiliency assurance services go hand in hand to support all kind of scenarios. And Yotta provides the flexibility and choice as per customer's requirement.

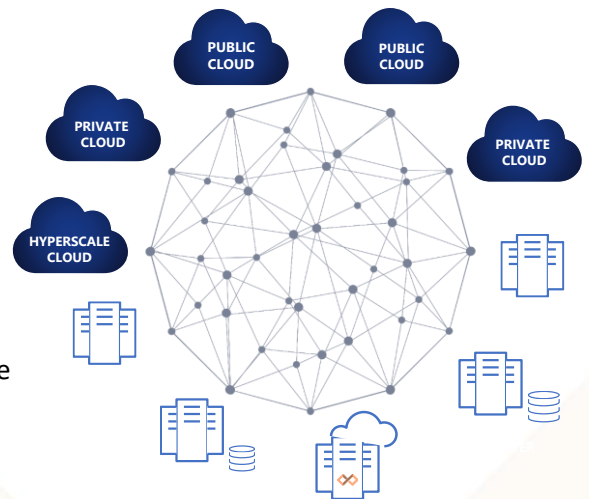
Sr No	Primary Site / DC	DR / Secondary Site Option		
1	DC @ Yotta Colo	DR @ Yotta	DR @ Customer Premise	DR @ Cloud
2	DC @ Yotta managed by Yotta	DR @ Yotta	DR @ Customer Premise	DR @ Cloud
3	DC @ Yotta Cloud	DR @ Yotta	DR @ Customer Premise	DR @ Cloud
4	DC @ Customer Premise	DR @ Yotta	DR @ Customer Premise	DR @ Cloud
5	DC @ Cloud	DR @ Yotta	DR @ Customer Premise	DR @ Cloud

Multiple options are possible but at same time we can segregate these scenarios into two broad categories.

- DC and DR both sites are managed by Yotta.
- Only the DR site is managed by Yotta.

Managing both DC and DR or only DR, gives Yotta complete visibility and control over the infrastructure and this enables monitoring and management of replication etc. That's why this enables complete deliveries to the customer.

But in the case of only DR / Secondary site is managed by Yotta, can offer the same services with little limitation and dependency on the customer. And those details are carved and penned down in the scope of the work document.



Service Features

RO consists of IT technologies and best practices designed to prevent or minimize data loss as well as business disruption resulting from catastrophic events like - everything from equipment failures and localized power outages to cyberattacks, accidents and natural disasters.

Sr No	Features Description	YOTTA RAS
1	Real-time Visibility of the entire IT environment including Business	✓
2	Service Availability Improved revenues through efficient Operations	✓
3	Assessing the impact of the Critical, Unplanned IT outage on the business financially	✓
4	Dashboard with real-time seamless information that empowers the stakeholders/ executives	✓
5	Automation of Business Services Continuity & Availability management	✓
6	Reduce Impact of IT Failure & Improve RTO/RPO for decision-making and recovery	✓
7	Auto-discovery of application infrastructure and application dependency	✓

Sr No	Features Description	YOTTA RAS
8	Single-click switchover/switchback	✓
9	Orchestration for all technology layers	✓
10	Heterogeneous IT environments supported	✓
11	Agent based and agent-less product, with multi-Tenancy on demand support	✓
13	Intelligently Automated Recovery Tests	✓
14	Integration and Automation for Network and Firewall	✓
15	Supports Physical, Virtual and Cloud Platforms	✓
16	Remote Management and Role-based User Access control	✓
17	Application-centric Recovery Management with end-to-end automation capabilities	✓
18	Central console for recovery management activities such as non-intrusive recovery drills for pre-flight checks, Switchover, Switchback, Failover and Failback; application start/ stop activities, etc	✓
19	Alerts on deviations in health parameters such as disk space, password changes, file addition/ deletion for prompt corrective actions	✓
20	Agentless monitoring of infrastructure health and custom application services	✓
21	Mitigation and the drastic reduction in recovery operational risks at a favorable price point	✓

Service Scope of Work

Sr No	Service Components	Details of Activities
1	Supported Environment	- Physical, virtual, VMware, Hyper V, KVM etc. <i>For more details, please refer "Technology compatibility matrix"</i>
2	Supported Replication	- Native Replication, Host-based / 3rd party replication - Carbonite – file-based and block-based Replication from Yotta, as per subscription - Storage replication for all branded OEM <i>For more details, please refer "Technology compatibility matrix"</i>
3	Implementation	1. Project Kick off Yotta will facilitate a kick-off meeting between the customer team and Yotta team on a basis of mutually agreed schedule. Yotta will: - discuss project plan in detail along with team's roles and responsibilities. - review the Services objectives. - review the Client environment, including required prerequisites specified by Yotta, and identify required OS environment upgrade and/or modification; and develop a preliminary schedule of activities. Customer will: - provide prerequisites (including system upgrades or modifications, environment access, and user privileges) specified by Yotta and documentation that describes Client's current environment; and - schedule and ensure participation of the subject matter experts to participate in the kick-off meeting.

Sr No	Service Components	Details of Activities
		2. Service Implementation <i>Assessment, Analysis and Workflow Development</i> The Yotta services specialist will work with customer SPOC to perform implementation planning activities which includes providing access to the Onsite configuration of Client RO Server (Customer site RO Server – server provided by customer and used solely for Resiliency Orchestration Software, if any of one side of site is not in Yotta DC). Yotta will conduct a planning meeting either F2F or over video conference basis the mutually agreed timelines with the customer team. Yotta will: - Analyse existing recovery procedures provided by customer, for completeness and readiness for conversion to automation workflows. If Yotta determines that existing procedures must be modified to automate, customer will be requested to provide the updated recovery procedures with the detailed changes to adjust and enable the recovery automation solution. - configure automation workflows on customer Resiliency Orchestration Server; and - Analyse and provide recommendations applicable for incorporation of the Services within customer’s environment. Customer will: - provide existing successfully tested (by a manual disaster recovery plan and drill) recovery procedures, recovery solution design, and Exercise documentation for the Services environment. - Be responsible for deployment and testing for User Acceptance Testing (UAT) environment. - Participate in the planning activities. - Ensure the validity of the existing disaster recovery and data replication environment, application and infrastructure dependencies and network infrastructure for the automation of those capabilities. - provide secure access to Client’s IT infrastructure at Covered Addresses necessary to allow Yotta and its subcontractors access and the ability to download and upload the required logs and install and configure Resiliency Orchestration Software and patches and necessary materials required to perform the Services; and - Provide access to Customer’s IT infrastructure at Covered Addresses beyond the Client firewalls as required by Yotta to perform the Services. Services Configuration Yotta will: - install and configure Resiliency Orchestration Software. - Configure automation workflows for the Services. - review the automation workflows with Client SPOC and - Configure the initial test of select components within the Services environment. - Customer will provide information required to perform the Services configuration.

Sr No	Service Components	Details of Activities
		Yotta will: - Lead and perform initial test of selected components for one application within the Services environment: and - Document and deliver to customer SPOC the Resiliency Orchestration Operation Guide that outlines necessary steps to recover designated components within the Services configuration. Customer will - Schedule and participate for the initial test and support the development of the Resiliency Orchestration Operation Guide. Completion Criteria: Yotta has: - Delivered the Resiliency Orchestration Operation Guide and done the briefing to the customer Point of Contact. - Performed an initial test of select components within the Services environment. Note: Completion of Services Implementation is deemed the Services Ready Start Date.
4	Administration	RAS User Account Management - Create /drop/alter RAS service account & User Accounts - Grant Access Rights to the user as per roles - Manage user profiles / Policies. RAS Administration - Monitor status of Orchestration dashboard - Monitor and manage Orchestration Server (upgrades, patches etc.) - Respond to orchestration alerts about health of recovery environment. - Perform problem determination and identify fix. - Perform any Preventive major as and when situation occur. - Review pre-test plans - Perform Pre-flight simulations. - Yotta team will do the monitoring and management of RO over internet VPN and client to provide Internet link (minimum of 2Mbps) for remote team to perform the monitoring and management. - Preventive major will be taken by RO remote team if in case of any issue or breach observed while doing regular monitoring and at the same time if it's found any issue pertaining to the client application/DB or other related platform, same will be reported to the customer as well. - Basis the number of drills and scope, regular Drills and DR Run Book will be performed by remote team apart from the day-to-day monitoring and management of RO.
5	Change & Configuration Management	Workflow /RPO/RTO Changes execution - Creation of workflow - Update existing workflow and configured RPO, RTO
6	Fault Management	Response to alerts generated by systems or problems reported Troubleshooting, RCA, and identification of problem area

Sr No	Service Components	Details of Activities
		Resolution of problems through configuration changes/ re-installations Restart / connectivity issues / Kill sessions / fix broken jobs or queues (Not for application)
7	Performance Management	24 X 7 monitoring of performance parameters (as per set metric or threshold) • Auto alerts and notifications to performance degradation/threshold Violation • Execution of Dynamic Troubleshooting & Performance Views and Script Identifying bottlenecks and suggesting improvement solution

Service Requisites

- Customers need to subscribe to the service as per subscription to avail RAS service.
- Basic IT infrastructure like, compute infra, internet or connectivity and data mover tools, which need to be subscribed or to be provided by the customer.
- Certain ports are to be opened from the customer side for collection of data and communications, for which customers will cooperate and share the relevant rights and details.
- Resiliency Orchestration Connectivity Requirements:

The RO Clients can work on agent / agent less in participation of nodes. Please find the requirements below:

- ✓ Network Access to the primary and DR servers from RO servers are necessary.
- ✓ Sufficient computer and storage space required for the installation of the binary.
- ✓ DR administrator name and password and 1 DR operator name and password must be provided at the time of configuring Yotta RAS RO. These are the users who will be using RO for admin and monitoring purposes. These users will be configured in RO software only.
- ✓ RO uses 45000, 45443, 46000 and 46001 TCP port for communication. Therefore, we need to open these ports if servers are behind firewall/DMZ. Additionally, solution specific ports also need to be opened.
- While provisioning service, delivery and assurance Team will share further any more info required or rights.

Please Note:

These are a few examples and details further can be shared by the project manager or technical account manager while deployment.

Following inputs are expected to be shared by the customer to deliver the optimized solution and service.

- Applications and IT environment landscape (please fill in the details while presales as well as while post deployment).

- If there is any readymade template of DR plan or workflow available.
- Or else please share step wise application Fail-Over / Recovery Method and its dependencies.
- Stepwise network Fail-Over / Recovery Method.
- Start-Up, Shutdown, Re-Start procedures of Applications, Database, etc.
- List of IT equipment & assets which needs to be considered for the project.
- Relevant communication matrix or SPOCs for Application, infra, DC and DR site perspective.
- Recovery Strategy and if any RPO and RTO expectations then discuss as well as fix while solutioning.
- I suggested to test before going live and make sure same continue as DR drill with a set frequency interval.

RACI Matrix

A	Pre-implementation Tasks	Yotta	Customer	Remarks
1	Data collection for in-scope application	RA	CI	System IT team and network team to work on pre-requisite at both sites
2	Share Yotta RAS pre-requisite	RA	I	
3	Provide hardware at DC and DR for RO installation	CI	RA	
4	Provide data for applications / servers as per templates	CI	RA	
5	Verify and configure pre-requisite with the customer	RA	RCI	
6	Conduct a meeting to walk through an implementation plan	RA	CI	
7	Sign Off on the plan	RA	R	Mutual
B	Onboarding Customer DB-Tier	Yotta	Customer	Remarks
1	RO configuration	RA	CI	DB team to configure and manage replication if yotta manages or else customer needs to do itself
1a	Database Management / Administration	CI	RA	
1b	Application Management / Administration	CI	RA	
1c	RPO Dependency (Link bandwidth, functioning of underlying data replication other than application binaries)	CI	RA	
2	DB replication Configuration	CI	RA	
3	RAS DB Workflow configuration	RA	CI	
4	RAS Monitor Job configuration	RA	CI	
5	RAS Alert configuration	RA	CI	
C	Onboarding Customer Application-Tier	Yotta	Customer	Remarks
1	Application installation & Configuration	CI	RA	Application team to install, and configure
2	Replication of application configuration, files	CI	RA	

3	RAS Application Workflow configuration	RA	CI	applications at both sites and share information RAS team for workflow configuration
4	RAS Monitor Job configuration and Alerts configuration	RA	CI	
D	Testing	Yotta	Customer	Remarks
1	Verification of App & DB workflow	RA	RA	Mutual
1a	Provide DR Invocation Request	CI	RA	
2	Conduct drill and test	RA	CI	
3	Verification of alerts and reports availability	RA	CI	
E	DR Drill and Go Live			
1	Test all the details and switch over and switch back	RA	CI	Mutual
2	Go live and share the RAS portal console with customer	RA	CI	

Key Assumptions

The following assumptions are hereby acknowledged by the parties and apply to the performance of this project under this SOW:

- Yotta RO implementation will be initiated only after the DR setup for application / nodes in scope is ready with one Time Sync and ongoing replication configured.
- Stake holders who are responsible for Administration / Management of Database, Application, Operating System, Network will provide necessary support.
- Delivery model for implementation will be if on-site and client will provide required Remote VPN client connectivity for remote managed services.

SUPPORTED TECHNOLOGY PLATFORMS

DR Orchestration (Perpetuuti – Continuity Patrol™)

OS Platforms	DB & Replication	Storage & Backup	Hypervisors	Cloud etc.
• Windows • RHEL • SLES • Cent OS • AIX	• MS SQL, Native Log Shipping • My SQL (Linux platform) • Oracle DG	• IBM SVC, Global Mirror • Dell EMC – SRDF, recover point, Mirror View • Hitachi Data System • HPE • NetApp – Snap Mirror • Veritas • Commvault	• Hyper-V • VMware • Citrix Xen • Oracle OVM • RHEV • HA VPAR • KVM	• AWS • AZURE • GCP • IBM SoftLayer • DNS Automation • NAT Automation • WebSphere • WebLogic • Tomcat & MS MQ

Carbonite Integration Overview – Windows

Carbonite Availability/Migrate Version

Carbonite Current Version: 8.5.8

Windows Server Version	Key Status	Support Highlight
Windows Server 2025	✓ Supported	Support for the latest OS release.
Windows Server 2022	✓ Supported	Current-generation long-term servicing channel (LTSC) OS.
Windows Server 2019	✓ Supported	Widely deployed previous generation OS.
Windows Server 2016	✓ Supported	Standard for many enterprise environments.
Windows Server 2012 R2	✓ Supported	Continues support for this popular version.
Windows Server 2012	✓ Supported	Maintains compatibility for this older environment.

Note: Cloud management services related to these technologies, please refer to the Technology Matrix available On Athena under the Cloud Assure Services section.

Carbonite 8.5.8 Integration Overview – Linux

OpenText Availability/Migrate Version

Carbonite Current Version: 8.5.8

Linux Distribution Family	Specific Versions Supported	Key Status
Red Hat Enterprise Linux (RHEL)	9.6, 9.5, 9.4	Comprehensive support for the latest major release branch (9.x).
CentOS	9.3	Supported, Note: No Oracle Linux (OL) support for 9.3.
Oracle Enterprise Linux (OEL)	8.10, 8.9, 8.8	Broad support for the 8.x major release branch.
Rocky Linux	8.7	Supported, note: Not supported with the unbreakable kernel.
	7.9, 7.8, 7.7	Support for the legacy 7.x major release branch.
SUSE Linux Enterprise Server (SLES)	15.6, 15.5, 15.4	Excellent support for the latest SUSE 15 service packs.
Ubuntu	24.04.2, 24.04.1, 24.04.0	Full support for the newest major LTS release (24.04).
	22.04.5	Supported version of the previous LTS release (22.04).
Debian Linux	11.4	Support for a specific stable release version.
Cloud Linux	7.9	Supported version for Cloud Linux environments.

Note: Cloud management services related to these technologies, please refer to the Technology Matrix available On Athena under the Cloud Assure Services section.

Service Plans

Service Plans	Plan Description	UOM
RESILIENCY ASSURANCE SERVICE (RESILIENCY ASSURANCE SERVICE PHYSICAL INSTANCE RO-WEB OR APP OR MIDDLEWARE)	RESILIENCY SERVICE ORCHAISTRATION AND AUTOMATION_PER PHYSICAL INSTANCE >Software subscription, Monitor as well as Management and ongoing steady state support of the service during the contract period is part of providers responsibility. >Physical Instance for WEB or APP and replication can happen through using inbuilt or native replication tools along with managed resiliency Orchestration and Portal > Customer IT infrastructure is not part of this, same can be subscribed separately from provider with extra charges	EA
Resiliency Assurance Service (RESILIENCY ASSURANCE SERVICE PHYSICAL INSTANCE RO-DB)	RESILIENCY SERVICE ORCHAISTRATION AND AUTOMATION_PER PHYSICAL INSTANCE >Software subscription, Monitor as well as Management and ongoing steady state support of the service during the contract period is part of providers responsibility. >Physical Instance of DB like MSSQL, Oracle and PostgreSQL >Replications happen through using inbuilt or native replication tools along with managed resiliency Orchestration and Portal > Customer IT infrastructure is not part of this, same can be subscribed separately from provider with extra charges	EA
Resiliency Assurance Service (RESILIENCY ASSURANCE SERVICE VIRTUAL INSTANCE RO-WEB OR APP OR MIDDLEWARE)	RESILIENCY SERVICE ORCHAISTRATION AND AUTOMATION_PER VIRTUAL INSTANCE >Software subscription, Monitor as well as Management and ongoing steady state support of the service during the contract period is part of providers responsibility. >Virtual Instance for WEB or APP and replication can happen through using inbuilt or native replication tools along with managed resiliency Orchestration and Portal > Customer IT infrastructure is not part of this, same can be subscribed separately from provider with extra charges	EA
Resiliency Assurance Service (VIRTUAL INSTANCE- DB)	RESILIENCY SERVICE ORCHAISTRATION AND AUTOMATION_PER VIRTUAL INSTANCE >Software subscription, Monitor as well as Management and ongoing steady state support of the service during the contract period is part of providers responsibility. >Virtual Instance of DB like MSSQL, Oracle and PostgreSQL >Replications happen through using inbuilt or native replication tools along with managed resiliency Orchestration and Portal >Customer IT infrastructure is not part of this, same can be subscribed separately from provider with extra charges	EA
Resiliency Assurance Service PHYSICAL INSTANCE - REPLICATOR	RESILIENCY SERVICE BLOCK LEVEL REPLICATION_PER PHYSICAL INSTANCE >This SKU enables and provides only Software subscription >Monitor, third party integration, automation, Orchestration as well as Management and ongoing steady state support of the service during the contract period, is part of the customer's responsibility. Same can be subscribed separately from provider with additional charges.	EA
Resiliency Assurance Service Virtual INSTANCE- REPLICATOR	RESILIENCY SERVICE BLOCK LEVEL REPLICATION_PER VIRTUAL INSTANCE - This SKU enables and provides only Software subscription - Monitor, third party integration, automation, Orchestration as well as Management and ongoing steady state support of the service during the contract period, is part of the customer's responsibility. Same can be subscribed separately from provider with additional charges.	EA
Resiliency Assurance Service INTEGRATION	RESILIENCY SERVICE INTEGRATION OF NATIVE OR 3RD PARTY TOOLS PER MANDAY >This service is applicable only based on FEASIBILITY CHECK and further decided as per reference of COMPACTIBILITY MATRIX, which is illustrated in detail, under PSS. >This Engineering or professional services for integration may require in case of native tools outside of the interoperability or Portal visibility and integration with data replication monitoring purpose e.g., native or 3rd party tools like Carbonite, Zerto, etc or any Hypervisor or Cloud or storage-based replication	EA

Resiliency DRILL Management Service	RESILIENCY SERVICE - DRILL MANAGEMENT_PER EVENT - Start-up Plan (1 To 5 instances)	EA
Resiliency DRILL Management Service	RESILIENCY SERVICE - DRILL MANAGEMENT_PER EVENT - Enterprise Plan (6 To 25 instances)	EA
Resiliency DRILL Management Service	RESILIENCY SERVICE - DRILL MANAGEMENT_PER EVENT - Enterprise Plus Plan (>25 instances)	EA
MIGRATION TOOL_ CARBONITE	Carbonite Migrate On-Demand tool for Windows and Linux per job or Per source side Instance. > Instance can be physical or Virtual and Maximum validity of License is 30 days from the applied date	EA

Pricing for the Resiliency Assurance Service includes both non-recurring charges and monthly recurring charges. Please refer to the Pricing Schedule or Tariff card for detailed rates.

- Non-Recurring Charges (NRC): Fixed Installation Fee / Service set up (installation, configuration) / integration and DR Drill or test
- Monthly Recurring Charges (MRC): The MRC is priced as per the type of RAS service and for regular activities, which is illustrated below.

Service Limitations

The following items are out of scope of this SoW:

- Service details and scope are clearly set out above. Other than this any additional infra, network, or any other services need subscription and same shall be chargeable.
- Installation, maintenance, monitoring or changing management of Customer's applications or databases. This might be offered as separate service and extra chargeable.
- Supply of RO license under this SOW, Supply, installation, upgrade or migration of any additional hardware, media, software, networking, and security equipment's, OS and other licenses outside the current scope.
- Implementation of data replication/ data mover solution must be factored under the separate scope and should be part of the overall solution.
- ✓ If native replication / 3rd party that needs to be considered.
- Any relocation cost of the Recovery system or equipment from the Production to the backup site.
- DR site readiness, network devices and bandwidth provisioning and installation of Hardware/software, except the implementation of DR Management Suite.
- Design or redesign of firewall or security policy.
- Performing any audits or examination for attacks or intrusions, vulnerability assessment, penetration tests, security parameters.

- Hardware support or break fix other than coordination and follow up with other 3rd party hardware vendors, who have supplied hardware directly to customer, as part of vendor coordination.
- Any Traffic analysis like Protocol wise traffic / Host wise traffic etc.
- Log Management / Log Analysis.
- Monitoring and fine tuning of any IDS / IPS / content filtering / Spam filtering / Data leakage / DRM functionality.
- Performing any infrastructure or application architecture consultancy.
- Automation of steps required for Fall Back in case actual disaster scenario.
- Initial full data synchronization between production and DR servers.
- Prepare or update Business continuity plan and Business Impact Analysis other than DR Run Book.
- Automation of tasks executed from Graphical User Interface using Yotta RO.
- Recovery Point Objective (RPO)
- Installation & configuration of Resiliency Orchestration at Backup site in case of Failover to DR site.
- All other deliverables that have not been specifically agreed and mentioned.
- Customer/Partner Responsibilities – DR Orchestration.
- Installation and Configuration of all hardware/OS/Software items except for Yotta RO software, on production and DR systems as per the Yotta RO software solution pre-requisites and specified DR solution configuration parameters.
- Customer will provide procedure / scripts for application automation.
- If UAT planned, the required installation and Configuration of all hardware/OS/Software as per the Yotta RO software solution pre-requisites and specified DR solution configuration parameters.
- Provisioning data replication bandwidth and last mile public or private connectivity as per requirements.
- Provisioning the required hardware required for the Yotta RO server if both the sites are out of Yotta DC.
- Customer will participate in Application Switchover / Switchback drill.
- Detail Procedures, Process Steps and information requested from Customer by Yotta RO as required to perform the Services outlined in this proposal must be accurate, complete and up to date (documented recovery procedures for the applications/ databases / systems under this project).
- Provide scripts for services, technologies under customer scope e.g application.

- Provision required disk space on DRM Server to create metadata backup and move to external Tape / disk media as per client Backup retention policy.
- Provide standard SMTP access for Email Alerts configuration from RO Portal.
- Provide recovery & drill procedure, which may include scripts or commands for the process or applications for which there is no direct support in Yotta Resiliency Orchestration.
- Customers will provide required access during the project implementation duration to allow Yotta RO team access and the ability to download and upload the required logs and Software program/patches and any other materials related to the Services to be performed under this SOW.
- Customer will meet all the Pre-requisites of provisioning hardware requirements for Implementing Yotta RO.
- Customer will take necessary steps to upgrade Operating System or Database instance if it is required to be supported by Yotta RO.
- Customers will provide Yotta with VPN connectivity to both DC and DR to enable Yotta team to perform this implementation remotely.

Value Propositions

Yotta Resiliency Assurance Service offers Disaster resiliency, automation, Orchestration, monitoring, reporting, testing, and workflow automation capabilities for complex hybrid IT environments in a scalable, easy-to-use solution. And its major advantages are.

Assert RPO-RTO SLAs



50-80%
Reduction in RTO
over manual / script
based recovery

Gain competitive edge through
increased visibility and
availability of business services

↑ Productivity



Anytime DR Drills
(Switchover-failover)
Scale & protect 100s of
applications with a
'Single Click'

Reduce cost and dependency
on SME's by avoiding weeks of
planning DR exercises

↓ Complexity



50+ %
Reduction in
Operational Cost

No need for plethora of siloed 3rd
party orchestration tools for
recovery of heterogeneous IT layers

✓ Compliance



Anytime BCP
Compliant Fully
automated drill reports

Save \$\$\$M by avoiding costly
Compliance penalties and lengthy
manual reporting procedures

Service Support

24x7 Support Desk provides a central point of contact between customers and our data center operations team. It encompasses a range of services reaching beyond the typical Help Desk including:

- Ability to process incidents and complaints.
- Process Change and Service Requests using IT service management processes i.e., Release Management, Configuration Management, and Problem Management etc.
- Call & Ticket Management functions.
- Our Support Desk encompasses a Call Centre and Help Desk Activities and hence shortens the incident life cycle efficiently.

Yotta has a strong team of highly skilled, experienced project managers who have implemented complex projects providing clients with value-added benefits such as risk management, clear communication, and greater visibility throughout the project life cycle.

- 99.95% Resiliency Assurance Service (RAS) infrastructure availability and uptime guarantee.
- Further SLA adherence is to the Service Response time commitments as mentioned in the table below. i.e., If the Actual Uptime during the month under consideration is less than this will lead to SLA violation and, Yotta shall provide Credits as set out herein.

SLA - Availability		Downtime per Month	Service Credits (%)
From (%)	To (%)		
99.95	<100	0 to 21.6 mins	0% of Monthly Recurring Charge (MRC)
99.94	<99.95	> 21.6 mins to 25.92 mins	10% of Monthly Recurring Charge (MRC)
99.93	<99.94	> 25.92 mins to 30.24 mins	15% of Monthly Recurring Charge (MRC)
< 99.90		> 43.20 minutes	25% of Monthly Recurring Charge (MRC)

Credits

Yotta offers comprehensive service level agreements for this service as uptime and Time to Response (TTR) to demonstrate its commitment to providing and maintaining the highest levels of service.

If in any month service availability is not fully available, the customer will be eligible for a Cash Credit equal to the percentage of the Monthly Recurring Charge (MRC) for the service affected by the lack of service availability subject to the limitations outlined in the Service Schedule and SLA. Any SLA credit relates to network availability, packet loss, latency etc, is covered under a specific service schedule and MSA relates to other services and its related SOF.

To receive a Cash credit, the customer shall open a trouble ticket via email to Provider NOC. In case the Customer

Is experiencing a violation of the committed Service Quality Parameter must open a trouble ticket within 10 (ten) Hours of such failure. The provider will use reasonable efforts to determine the source of such outage and to correct Such problem to the extent that the source of the problem is on the Provider network or setup.

Credits will generally be reflected on the second invoice following the billing month in which the Service-affecting event occurs. The credits provided in this service specification sheet are the customer's sole and exclusive remedies for all matters related to the guaranteed criteria.

Yotta agrees to supply the service as per the terms of this agreement. If the Supplier needs to conduct non-Emergency maintenance work outside of a scheduled regular maintenance Window, it will do so after having provided the Customer with seven (7) days of written notice about the work and the potential for the interruption to the Service.

Yotta shall not be obligated to inform Customer in advance of Emergency Maintenance work.

Service Exclusions

The following conditions are specifically excluded from coverage under this SLA.

- Force majeure events or customer's failure to materially comply with its obligations as defined in agreement; or,
- An interruption to service previously notified in writing, 7 calendar days before, to the Customer by Provider as a part of scheduled maintenance, reconfiguration, or testing activities: or,
- Any components operated by an Internet Service Provider or a network operator other than Provider; or,
- Troubles resolved as "No Fault Found"; or,
- Failure of local access used to access the Yotta network; or,
- Outages less than 60 seconds in duration; or,
- Time attributed to Customer's delay in responding to Provider' requests for assistance to repair an outage; or,
- Failure of any components beyond Yotta's control or caused by factors outside of Yotta's reasonable control; or,
- That resulted from any actions or inactions of the customer, or any third parties engaged by the Customer.
- That resulted from the Customer's equipment and/or third-party equipment, not within the sole control of YOTTA.
- Any restrictions placed/imposed upon YOTTA by the Customer, which prevent or delay the performance of YOTTA's obligations including but not limited to restrictions on YOTTA's right to access the Designated Customer Area resulting in YOTTA's failure to perform necessary maintenance or emergency / supplemental work in time.
- Service suspension (under MSA) in accordance with terms of the Agreement.

Incident Management

Yotta will investigate all faults that get captured proactively on the system using monitoring tools and will manage Till the resolution of the incident. The response and resolution SLA for various incident categories are tabulated below:

Incident Type	Response	Update	Resolution
S1 – CRITICAL - Complete Service Outage Critical (Total Non-availability of service) - Incident causing disruption by impacting on customer's critical business functions and need immediate attention	30 Mins	60 Mins	4 Hours
S2 – HIGH - Partial Outage Major (Group of users are affected) - Incident causing disruption by impacting on customer's critical business functions and need attention	60 Mins	180 Mins	8 Hours
S3 – MEDIUM - Partial Performance impact Minor (Performance issues, Single user issues) -Incident causing degradation of service	90 Mins	360 Mins	36 Hours
S4 - Request for Service or Information	90 Mins	360 Mins	36 Hours

Problem Management

Incident trend analysis will be tracked through problem tickets as appropriate for arresting recurring incidents. Known Error Database and Knowledge articles will be maintained for reference during product life cycle management.

Change Management

Customer requests for changes related to service are tracked through the service requests and are received by the Support Desk. A change ticket is generated for each change request and tracked for closure through feasibility, Impact analysis, implementation, and post-implementation review. A list of emergency and standard change requests allowed for the service is listed below. This excludes any delay in correspondence from the customer.

Change Type	TAT for Request Fulfilment
Standard Change	1 Business Days
Emergency Change	4 hours from receiving the emergency change request, within a calendar Business Day

Configuration Management

Yotta refers to the Configuration Management Database (CMDB) to determine the best course of action when the service needs repair, modification, expansion, or upgradation. This database contains the locations as well as information about the hardware configuration, versions, and installed patches. This helps.

- Streamlining the processes of maintenance, repair, expansion and upgrading
- Minimizing configuration errors
- Minimizing downtime
- Optimizes security postures.
- Ensuring that changes made to a device do not adversely affect other devices or systems.
- Rolling back changes to a previous configuration if results are unsatisfactory.
- Archiving the details of all configuration changes

Software Patch Management: Patches and updates to the Yotta infrastructure are internally planned and executed As planned maintenance activity. Yotta performs scheduled maintenance on weekends, such that normal business Activity is not disrupted. When any planned maintenance activity could affect the customer's service, an email notification describing the maintenance activities and any expected service downtime (downtime is required for activities like firmware upgrade or controller head swap) are sent to the customer at least in advance as per Master Service Agreement (MSA) standard.

Yotta also reserves the right to perform any unplanned emergency maintenance as needed without prior Notification to the customer. However, if emergency maintenance requires any service interruption, then customers are notified by email with relevant information.

Note: In case the customer does not agree to deploy any of the essential/mandatory patch/firmware/software deployment against OEM / Vendor recommendations, Yotta will continue to support such customer service on a best -effort SLA. This best-effort SLA will be applicable from the date Yotta communicates to the customer about such patch update recommendations.

Reporting: As a part of standard reporting functionality, the following reports will be sent to the customer or Available under the customer portal as a part of the service. Any other report may be considered or offered on a need basis with additional cost and subject to feasibility.

- List of Incident / Service Requests/ Change Requests for a month
- Performance and SLA

Service Assurance

- The customer can reach our support desk over a telephone or e-mail and seek a case number (Trouble Ticket - TT)
- Support Email id – helpdesk@yotta.com
- Toll-Free numbers – 1-800-20-96882
- Within 30 minutes the customer will receive first-hand information about his case number.
- As per the severity of the problem it will follow the escalation till its resolution.
- In case where the Customer encounters Service Outage, the following escalation procedure shall be followed.
- If the call is not resolved within a specified time by Yotta, then the customer may intimate and escalate as per the escalation matrix shared below.

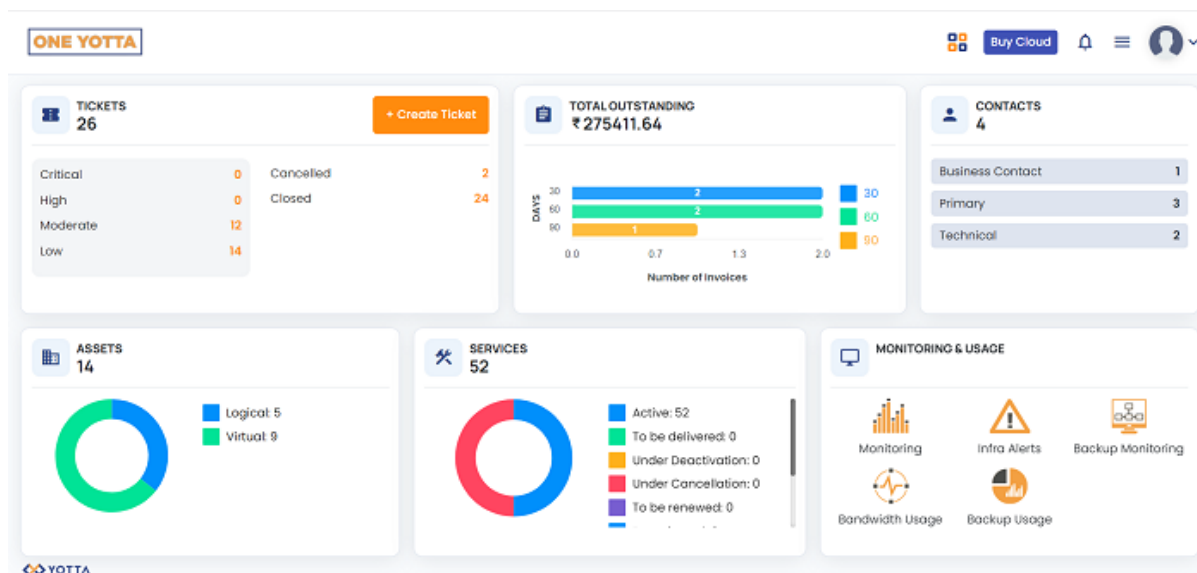
Escalation Matrix

Escalation level	Contact Name	Contact Information	Contact Email
Level 1	Helpdesk Team	1-800-20-96882	helpdesk@Yotta.com
Level 2	Shift In-charge	1-800-20-96882	shiftincharge@yotta.com
Level 3	Head Service Assurance	+91 8451847845	arevankar@yotta.com

Customer Portal

ONEYOTTA is our customer portal which can be accessed <https://account.yotta.com/>

This simplifies IT performance monitoring by displaying your systems health, utilisation data and metrics with real-time KPIs on a centralized single dashboard. Further with our specialists and experts we do daily maintenance and 24*7 prescriptive monitoring, so that seamlessly we identify, react, and resolve issues before hand it operationally impacts.



ONEYOTTA Features:

- **User Management:** Assign users and assign rights as per your policies.
- **Raise Tickets:** Raise and monitor any Incident and Service request for seamless and efficient tracking.
- **Billing Information:** Get all the billing information related to your account with a click: view Invoices, Credit Notes and Transactions.
- **View Assets:** Get complete visibility on your IT assets, Monitor data centre parameters, Network and Security systems, your server, storage and other assets.
- **View Services:** Single view for all your subscribed services. Get details on your active, expired, deactivated and all types of services.
- **Monitoring & Usage:** Monitor and view your Infrastructure, Backup and Usage details via a single dashboard.

ONE-YOTTA Benefits:

- Increase IT support efficiency, improve system performance as well as availability.
- Gain greater visibility into the health and utilization of enterprise IT systems.
- Due to Visibility, better routine maintenance and faster troubleshooting issues.
- Reduce downtime and unplanned outages.
- Improve ITSM quality and Enhance compliance.
- Lower risk of error and cost overruns
- Make smarter decisions with objective measurements for every KPI.
- Greater Service Level alignment & Increased business agility

Note: For more information, please visit www.yotta.com or contact our account manager.